FIITJEE

ALL INDIA TEST SERIES

PART TEST – I

JEE (Advanced)-2020

PAPER – 2

TEST DATE: 10-11-2019

ANSWERS, HINTS & SOLUTIONS

Physics

PART – I

SECTION – A

1. A, B, C, D

Sol. By definition

2. A, C

Sol. To find F draw the F.B.D. of the mass. Then apply work energy theorem to get maximum speed.

3. B, C, D

Sol. The number of collisions will be dependent on the masses of the three objects. If m_1 = 4m, m_2 = m and m_3 = 3m then m_1 finally moves to left with 1 m/s, m_2 with speed 1.5 m/s and m_3 to right side with speed 2.5 m/s.

4. A, D

Sol. Write the co-ordinates of the collar and differentiate w.r.t. time to get velocity of collar.

5. A, B, C

Sol. Let the line joining the centres of sphere and grindstone be inclined at ϕ to the vertical, and let R, F be the normal reaction and friction. The sphere remains at rest if:
$R = Mg \cos \phi$ and $F = Mg \sin \phi$

Also : $\frac{2Mb^2}{5}\frac{d^2\theta}{dt^2} = Fb = Mgb\sin\phi \Rightarrow \frac{bd^2\theta}{dt^2} = \frac{5g\sin\phi}{2}$

and $bd^2\theta/dt^2$ = acceleration of point of contact = $a\alpha$, therefore $\alpha = \frac{5g\sin\phi}{2a}$, so that α must not exceed $5g/2a$.

6. A, D

Sol. If $a = \frac{v_0}{t} > \mu g$, then the distance covered by the object relative to train

$$\ell = \frac{1}{2}(a-\mu g)t^2 + \frac{(a-\mu g)^2 t^2}{2\mu g}$$

$$= \frac{v_0 t}{2} - \frac{1}{2}\mu g t^2 + \frac{(v_0 - \mu g t)^2}{2\mu g}$$

$$= \frac{v_0^2}{2\mu g} - \frac{v_0 t}{2}$$

If the wagon long enough, distance moved by the small body relative to train is $\ell = \frac{v_0^2}{2\mu g} - \frac{v_0 t}{2}$

7. A, B, C

Sol. Let θ be the inclination of the direction of projection to the horizontal, and x the distance from the wall to the point of projection, so that the time to the wall = $x/v\cos\theta$. After the impact the horizontal velocity is $ev\cos\theta$, so that the particle will be vertically over the starting point again in time $x/ev\cos\theta$ from the impact .(e is the coefficient of restitution)
The vertical velocity is unaltered by the impact, so that the particle will be on the same vertical level as the point of projection in time $2V\sin\theta/g$.
The particle will therefore return to the point of projection if

$$\frac{2v\sin\theta}{g} = \frac{x}{v\cos\theta} + \frac{x}{ev\cos\theta}$$

$$\Rightarrow \qquad x = \left(\frac{e}{1+e}\right)\frac{v^2\sin 2\theta}{g}$$

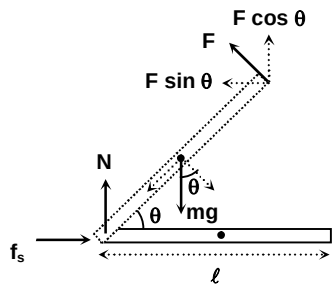
Now the greatest value of $\sin 2\theta$ is unity, so that the greatest value of x is $\frac{v^2 e}{g(e+1)}$, where e is the coefficient of restitution. Since $0 \le e \le 1 \Rightarrow x \le 5$

8. C, D

Sol. Since the rod is raised slowly (quasi-statically), the entire system remains in equilibrium at any moment of time. Thus, the torque about the point of rotation of the rod is zero about any axis, and the net external forces are 0.

$$\tau = -F\ell + \frac{mg\ell\cos\theta}{2} = 0 \Rightarrow F = \frac{mg\cos\theta}{2}$$

$N = mg - F\cos\theta$, $f_s = F\sin\theta$



The F sin θ force causes the rod's bottom end to slip in one direction. The friction force counters this force to stop the slipping, $f_s \le \mu_s N$

$\mu_s N \ge F \sin\theta$. Hence,

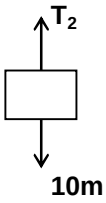
$$\mu_s \ge \frac{F\sin\theta}{mg - F\cos\theta} = \frac{\sin\theta\cos\theta}{2-\cos^2\theta}$$

Maximum of μs can be found by setting its derivative with respect to theta equal to zero:

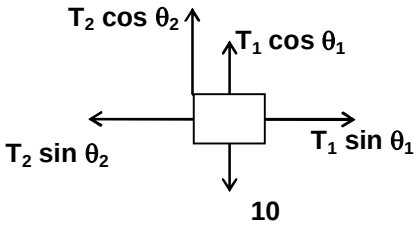
$$\frac{d\mu}{d\theta} = \frac{\left(2-\cos^2\theta\right)\left(2\cos^2\theta - 1\right) - 2\cos^2\theta\left(1-\cos^2\theta\right)}{\left(2-\cos^2\theta\right)^2} = 0 \text{ This gives } \sin\theta = \frac{1}{\sqrt{3}} \text{ and } \mu_{min} = \frac{\sqrt{2}}{4}$$

9. A

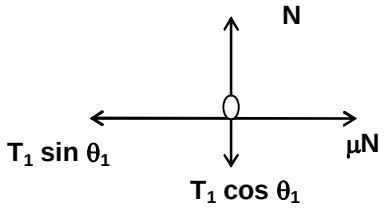
Sol.



F.B.D. of m



F.B.D. of 1 kg



F.B.D of ring

For ring

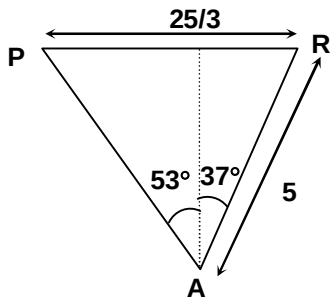
$N = T_1 \cos\theta_1$

$\frac{3}{4}N = T_1 \sin\theta_1$

$\tan\theta_1 = 3/4 \Rightarrow \theta_1 = 37°$

from geometry of triangle $\theta_2 = 53°$

$\Rightarrow \theta = 90°$



For 1 kg block

$T_1 \sin\theta_1 = T_2 \sin\theta_2 \Rightarrow 3T_1 = 4T_2$

$T_2 \cos\theta_2 + T_1 \cos\theta_1 = 10$

$\Rightarrow T_2 = 6$ N, $T_1 = 8$ N

For m

$T_2 = 10\ m = 6$

$m = \frac{6}{10}\text{kg} = 0.6$ kg

10. D

Sol. For 1 kg block

$T_1 = 8$ N

$T_2 - 6 = a$ ….(i)

Solving (i) and (ii), we get

$a = 6\ m/s^2$

$\therefore T_2 = 6 + a = 6 + 6 = 12$ N

For Block A

$5mg - T_2 = 5ma$

$30 - T_2 = 3a$ …(ii)

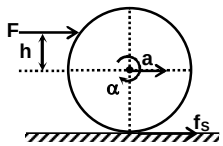
11. A

12. C

Sol. (for Q.11-12)

If $h > h_0$ $\left(\text{where } h_0 = \frac{I_{CM}}{mR}\right)$

$F + f_S = ma$ …(i)



$Fh - f_S R = I_{CM}\alpha$

$\frac{Fh}{R} - f_s = \frac{I_{cm}a}{R^2}$ …(ii)

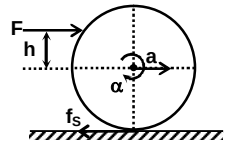
From (i) and (ii)

$F\left(1+\frac{h}{R}\right) = ma\left(1+\frac{I_{CM}}{mR^2}\right)$

$a = \frac{F\left(1+\frac{h}{R}\right)}{m\left(1+\frac{I_{CM}}{mR^2}\right)}$

$f_S = ma - F$

If $h < h_0$



$F - f_S = ma$ …(i)

$Fh + f_S R = I_{CM}\alpha$

$\frac{Fh}{R} + f_s = \frac{I_{cm}a}{R^2}$ …(ii)

From (i) and (ii)

$F\left(1+\frac{h}{R}\right) = ma\left(1+\frac{I_{CM}}{mR^2}\right)$

$a = \frac{F\left(1+\frac{h}{R}\right)}{m\left(1+\frac{I_{CM}}{mR^2}\right)}$

$f_S = F - ma$

For a ring,

$F = 2mg, h = R/2$

$a_0 = \frac{3g}{2}, f_0 = f_S = F - ma = \frac{mg}{2}$

$a_0 = \frac{3g}{2}, \ f_0 = \frac{mg}{2}$

For a disc,

$F = mg, h = 3R/4$

$a = \frac{mg\left(1+\frac{3}{4}\right)}{m\frac{3}{2}} = \frac{7g}{4}\times\frac{2}{3} = \frac{7g}{6}$

$f_S = ma - F = \frac{7mg}{6} - mg = \frac{mg}{6}$

$a = \frac{7a_0}{9}, \ f_S = \frac{f_0}{3}$

For a hollow sphere,

$F = \frac{5mg}{8}, h = \frac{R}{3}$

$a = \frac{\frac{5mg}{8}\times\frac{4}{3}}{\frac{5}{3}m} = \frac{g}{2} \ \therefore a = a_0/3$

$f_S = F - ma = \frac{5mg}{8} - \frac{mg}{2} = \frac{mg}{8} = \frac{f_0}{4}$

$f_S = f_0/4$

For a solid sphere

$F = mg/2$, $h = 3R/5$

$a = \frac{\frac{mg}{2} \times \frac{8}{5}}{m\frac{7}{5}} = \frac{4g}{7} = \frac{8}{21}a_0$

$f_S = ma - F$

$f_S = \frac{4mg}{7} - \frac{mg}{2} = \frac{mg}{14}$

$a = \frac{8a_0}{21}$, $f_S = \frac{f_0}{7}$

SECTION – C

13. 00002.25

Sol. $m\frac{dv}{dt}v = P$ = constant

When acceleration reduces to half value its speed becomes doubled i.e. $v = 2v_0$

$Pt = \frac{1}{2}m\left(v^2 - v_0^2\right)$

$Pt = \frac{1}{2}m\left(4v_0^2 - v_0^2\right)$

$\Rightarrow t = \frac{3mv_0^2}{2P} = \frac{3\times1\times100\times15}{2\times1000}$ = 2.25 sec.

14. 00001.20

Sol. $M_0 g\sin\theta + f_s = M_0 a$ …(i)

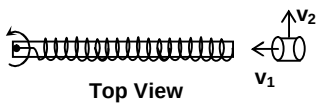
$f_s = Mg\sin\theta$ …(ii)

$a = \frac{(M_0 + M)g\sin\theta}{M_0}$

The time in which he gets to the other end of the plank is $\sqrt{\frac{2M_0\ell}{(M+M_0)g\sin\theta}}$

15. 00003.60

Sol. For entering without jerk $v_2 = \ell\omega$ = 2 m/s



Using work energy theorem on sleeve after entering in the frame of rod.

$W_{spring} + W_{centrifugal} = \Delta K$

$-\frac{1}{2}k\ell^2 - \frac{1}{2}m\omega^2\ell^2 = 0 - \frac{1}{2}mv_1^2$

$\Rightarrow \frac{1}{2}mv_1^2 = \frac{1}{2}k\ell^2 + \frac{1}{2}m\omega^2\ell^2$

Now $K = \frac{1}{2}m\left(v_1^2 + v_2^2\right) = \frac{1}{2}k\ell^2 + \frac{1}{2}m\omega^2\ell^2 + \frac{1}{2}m\omega^2\ell^2$

= 2 + 1.6 = 3.6 Joule

16. 00005.00

Sol. When the wedge has moved through a distance y, let the point of contact of cylinder have moved down the plane through x (with respect to wedge). Let F be the friction between the cylinder and the plane.

Since there is no horizontal force on (wedge + cylinder system)

$$M\frac{d^2y}{dt^2} + m\left(\frac{d^2y}{dt^2} - \cos\alpha\frac{d^2x}{dt^2}\right) = 0 \qquad \ldots(1)$$

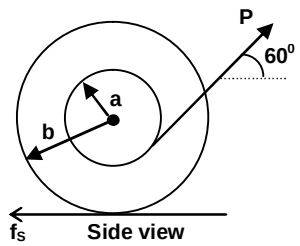
Also, $m\left(\frac{d^2y}{dt^2}\cos\alpha - \frac{d^2x}{dt^2}\right) = F - mg\sin\alpha$ and $\frac{ma}{2}\frac{d^2x}{dt^2} = \frac{ma^2}{2}\frac{d^2\theta}{dt^2} = Fa$, so that

$$2\frac{d^2y}{dt^2}\cos\alpha - 3\left(\frac{d^2x}{dt^2}\right) = -2g\sin\alpha \qquad \ldots(2)$$

Equation (1) and (2) implies acceleration of the centre of the cylinder down the face, and relative to the wedge, is $\frac{2g\sin\alpha(M+m)}{3M+m+2m\sin^2\alpha}$ = 5 m/s^2.

17. 00005.00

Sol. Assume that the drum rolls away from the force P, so that the friction F is away from P. Then :



$$Mk^2\frac{d^2\theta}{dt^2} = Pa - Fb \text{ and } Mb\frac{d^2\theta}{dt^2} = M\frac{d^2x}{dt^2} = F - P\cos 60°$$

$$M\left(k^2 + b^2\right)\frac{d^2\theta}{dt^2} = P\left[a - \frac{b}{2}\right]$$

$$\Rightarrow 0.25\times 8\frac{d^2\theta}{dt^2} = 10(1.5 - 1)$$

$$\Rightarrow \frac{d^2\theta}{dt^2} = 2.5 \text{ rad/s}^2$$

And the required acceleration = $b\frac{d^2\theta}{dt^2} = 2 \times 2.5 = 5$ m/s^2

18. 00012.00

Sol. $v_0\cos 30° = v\cos 45°$

$$\Rightarrow v = v_0\sqrt{\frac{3}{2}}$$

Now using conservation of energy

$$\frac{1}{2}m(v^2 - v_0^2) = mgh$$

$$\frac{3v_0^2}{2} - v_0^2 = 2gh \qquad \Rightarrow \frac{v_0^2}{2} = 2gh$$

$$\Rightarrow v_0 = \sqrt{4gh} = \sqrt{4\times 10\times 3.6} = 12\,\text{m/s}$$

Chemistry

PART – II

SECTION – A

19. A, B, D

Sol. Cathode : H_2, Anode : Cl_2

20. A, B, C

Sol. (D) : LiCl

21. A, B, D

22. A, B, C

Sol. $Mg = [Ne]3s^2$

3 + 0 + 0 = 3

23. A, B, C

Sol. $N_2O_4 \rightleftharpoons 2NO_2$

4 8

$K_P = \frac{8 \times 8}{4} = 16$ atm

On doubling the volume

$N_2O_4 \rightleftharpoons 2NO_2$

$\frac{4}{2} - x$ $\frac{8}{2} + 2x$

$16 = \frac{(4 + 2x)^2}{(2 - x)}$

x = 0.4725

24. A, B, C

25. B, C, D

Sol. (A) $CaC_2/Mg_2C_3 + H_2O \rightarrow C_2H_2 / C_3H_4$

(B) $Be_2C / Al_4C_3 + H_2O \longrightarrow CH_4$

(C) R_2SiCl_2

(D) In pyrosilicate one oxygen atom is shared.

26. B, D

Sol. n-Factor of H_3PO_2 is one

$200 \times \frac{1}{3} \times 1 \neq 100 \times 1 \times 2$

$pH = 7 + \frac{1}{2} \times 1.2 + \frac{1}{2} \log \frac{1}{2} = 7.45$

27. A

28. B

29. C

30. B

SECTION – C

31. 00001.50

Sol. $S \longrightarrow K_2SO_4 + 6e^-$ … (1)

$Mn(NO_3)_2 + 2e^- \longrightarrow K_2MnO_4 + 2NO$ … (2)

$3Mn(NO_3)_2 + S \longrightarrow K_2SO_4 + 3K_2MnO_4 + 6NO$

Now balance K by adding 8 KOH on left side.

$3Mn(NO_3)_2 + S + 8KOH \longrightarrow K_2SO_4 + 3K_2MnO_4 + 6NO + 4H_2O$

$= \frac{x + y + z}{10} = \frac{3 + 8 + 4}{10} = 1.5$

32. 00006.50

Sol. $PCl_3 + 3H_2O \longrightarrow H_3PO_3 + 3HCl$

$0.5 \qquad 0.5 \quad 3 \times 0.5$

No. of mole of NaOH = 0.5 × 2 + 3 × 0.5 × 1

= 2.5

$PCl_5 + 4H_2O \longrightarrow H_3PO_4 + 5HCl$

$0.5 \qquad 0.5 \quad 0.5 \times 5$

Number of moles of NaOH required = 0.5 × 3 + 0.5 × 5 × 1

= 4

Hence, total mole of NaOH = 2.5 + 4

= 6.5 mol

33. 00000.20

Sol. pH = 5

$[H^+] = 10^{-5}$

$RH_2 + Ca^{+2} \longrightarrow RCa + 2H^+$

$\frac{1}{2} \times 10^{-5} \qquad 10^{-5}$

Wt. of Ca^{+2} in one litre $= \frac{1}{2} \times 10^{-5} \times 40 = 2 \times 10^{-4}$ gram = 0.2 mg

Hence, hardness will be 0.2 ppm.

34. 00019.00

Sol. $2NaOH + CO_2 \longrightarrow Na_2CO_3$

20 mm 1 mm 0

18 mm 0 1 mm

meq. of $NaOH + \frac{1}{2}$ meq. of Na_2CO_3 = meq. of HCl

$18 \times 1 + \frac{1}{2} \times 1 \times 2 = x \times 1,$

x = 19

35. 00000.20

Sol. $A \rightleftharpoons B + C$

$\alpha = \frac{D - d}{d}$

$= \frac{50 - 41.67}{41.67} = 0.2$

$A \rightleftharpoons B + C$

1 0 0

$(1-\alpha)$ $1 \times \alpha$ $1 \times \alpha$

Hence number of mole of

$= 1 \times 0.2 = 0.2$

36. 00010.45

Sol. $\frac{49 \times \frac{1}{5} \times 1000}{40} + \frac{49 \times \frac{4}{5} \times 1000}{49} = V \times \frac{1}{10}$

$V = 10450 \text{ ml} = 10.45 \text{ litre}$

Mathematics

PART – III

SECTION – A

37. A, B, C, D

Sol. $\because\ 0 \le \int_0^1 (f(x)-x)^2\,dx = \int_0^1 f^2(x)\,dx - 2\int_0^1 xf(x)\,dx + \int_0^1 x^2dx$

So, $\int_0^1 f^2(x)\,dx \ge 2\int_0^1 xf(x)\,dx - \int_0^1 x^2dx = 2\int_0^1 xf(x)\,dx - \frac{1}{3}$

$\because\ \int_0^1 \left(\int_x^1 f(t)\,dt\right)dx \ge \int_0^1 \left(\frac{1-x^2}{2}\right)dx$

$\Rightarrow \int_0^1 tf(t)\,dt \ge \frac{1}{3}$

So, $\int_0^1 f^2(x)\,dt \ge \frac{1}{3}$

38. A

Sol. Putting $x^{\frac{1}{k}} = t$ gives $\int_0^1 (f(t))^{n-k}\ t^{k-1}\,dt = \frac{1}{n}\ \forall\ k \in \{1, 2, 3, ..., n-1\}$

So, $\int_0^1 (f(t)-t)^{n-1}\,dt = \int_0^1 \sum_{k=0}^{n-1} {}^{n-1}C_k\,(-1)^k\,(f(t))^{n-1-k}t^k\ dt$

$= \int_0^1 \sum_{k=1}^{n} {}^{n-1}C_{k-1}(-1)^{k-1}(f(t))^{n-k}t^{k-1}\,dt$

$= \sum_{k=1}^{n}(-1)^{k-1}\left({}^{n-1}C_{k-1}\right)\int_0^1 (f(t))^{n-k}\cdot t^{k-1}dt$

$= \sum_{k=1}^{n}(-1)^{k-1}\left({}^{n-1}C_{k-1}\right)\frac{1}{n} = \frac{1}{n}(1-1)^{n-1} = 0$

But $(f(t)-t)^{n-1} \ge 0$ as $n-1$ is an even positive integer

$\therefore\ f(t) = t\ \forall\ t \in [0, 1]$

39. A, B, C, D

Sol. $f''(x)\,f'''(x) \le f''(x)\,f(x) < f''(x)\,f(x) + (f'(x))^2\ \forall\ x \in R$

$\Rightarrow \frac{1}{2}(f''(x))^2 < f(x)f'(x)\ \forall\ x \in R\ \left(\because \lim_{x\to-\infty}\frac{1}{2}(f''(x))^2 - f(x)f'(x) = 0\right)$

$\frac{1}{2}(f''(x))^2 + \frac{1}{2}f'(x)f'''(x) < f(x)\cdot f'(x) + \frac{1}{2}f'(x)f''(x) < f(x)\cdot f'(x) + \frac{1}{2}f'(x)f(x)$

$\frac{1}{2}f'(x)f''(x) < \frac{3}{4}f^2(x)\ \forall\ x \in R \Rightarrow \frac{1}{2}(f'(x))^2 f''(x) < \frac{3}{4}f^2(x)f'(x)\ \forall\ x \in R$

$\Rightarrow \frac{1}{6}f'^3(x) < \frac{1}{4}f^3(x)\ \forall\ x \in R \Rightarrow f'(x) < \left(\frac{3}{2}\right)^{\frac{1}{3}} f(x)\ \forall\ x \in R$

40. A, B, C

Sol. $\frac{y\cdot y''-(y')^2}{(y')^2}=\frac{(2x-1)(y')^2}{(y')^2}$; $-\int d\left(\frac{y}{y'}\right)=\int(2x-1)dx$

$y=e^{\frac{4x}{2x-1}}$

41. A, B, C, D

Sol. $\int_0^1\left(x-\frac{1}{2}\right)f'(x)dx=\frac{f(1)+f(0)}{2}-\int_0^1 f(x)dx$

Since, $\left|\int_0^1\left(x-\frac{1}{2}\right)f'(x)dx\right|\le\max_{0\le x\le1}|f'(x)|\int_0^1\left|x-\frac{1}{2}\right|dx\le\frac{1}{4}\max_{0\le x\le1}|f'(x)|$

42. A, B, D

Sol. $\frac{e^x+e^{-x}}{2}=\sum_{n=0}^{\infty}\frac{x^{2n}}{\underline{|2n}}\le\sum_{n=0}^{\infty}\frac{x^{2n}}{2^n\underline{|n}}=e^{\frac{x^2}{2}}\le e^{cx^2}\ \forall\ x\in R\Rightarrow c\ge\frac{1}{2}$

For $c=\frac{1}{e}$, inequality does not hold for x = 1

43. A, B

Sol. (A) Clearly function is differentiable at some of these points
(B) Differentiable $\forall\ x\in[0,4]$
(C) Not differentiable at x = 2, 5, 8
(D) f(x) is not differentiable at 1

44. A, C

Sol. $\because\ \int_0^a f(x)dx=0$

Put x = at, we get $\int_0^1 f(at)dt=0$

Let $M=\max_{0<x<1}|f'(x)|$

Using LMVT on [a, ax], |f(x) – f(ax)| ≤ (1 – a)xM

Now, $\left|\int_0^1 f(x)dx\right|=\left|\int_0^1(f(x)-f(ax))dx\right|\le(1-a)M\int_0^1 xdx\le\frac{1}{2}(1-a)M$

Equality holds if $f(x)=\pm M\left(x-\frac{a}{2}\right)$

45. D

46. B

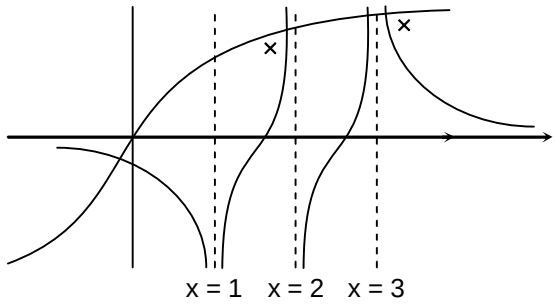
Sol. (45.-46.)
$f_1(x)$ is discontinuous at x = –1 only

$f_2(x)$ is continuous $\forall\ x \in R$
$f_3(x)$ is continuous at x = 0, $\sqrt{3}$ only
$f_4(x)$ is continuous at each irrational number

47. A

48. A

Sol. (47.-48.)
(I) $(x^3 - x + 1)^3 - x^3 + x = 0$
$\Rightarrow f(f(x)) = 0$, where $f(x) = x^3 - x + 1$
which has only one negative solution
(II) $f(x) = x^6 - x^4 + 2x^3 - x^2 + 1$
$= (x^2 - 1)^2(x^2 + 1) + 2x^3 > 0\ \forall\ x > 0$
(III) $\frac{1}{x-1} + \frac{1}{x-2} + \frac{1}{x-3} = x^{\frac{3}{5}}$ has three positive solutions
(IV) $2^x = x^4$, has two positive solutions



SECTION – C

49. 00011.00

Sol. f(3) = 8 and f′(0) = 3

50. 00001.00

Sol. $f(x) = f(0) + f'(0)x + \frac{f''(0)}{2!}x^2 + \frac{f'''(\alpha)}{3!}x^3$ where $0 \le \alpha \le x$ and

$f(x) = f(2) + f'(2)(x-2) + \frac{f''(2)}{2!}(x-2)^2 + \frac{f'''(\beta)}{3!}(x-2)^3$ where $x \le \beta \le 2$

$\frac{f'''(\alpha)}{6}x^3 = \frac{f'''(\beta)}{6}(x-2)^3 + 2$

Put x = 1 ; $f'''(\alpha) + f'''(\beta) = 12$

So, at least one of $f'''(\alpha)$ and $f'''(\beta)$ is ≥ 6

51. 00000.00

Sol. $\lim\limits_{x\to\infty} f(x) = \frac{f(x)\cdot e^{\frac{x^2}{2}}}{e^{\frac{x^2}{2}}} = \lim\limits_{x\to\infty} \frac{f''(x) + 2xf'(x) + (x^2+1)f(x)}{(x^2+1)}$

So, $\lim\limits_{x\to\infty} f(x) = 0$

52. 00000.00

Sol. Put $g(x) = x^2 + \frac{1}{4}$

$x \in \left[-\frac{1}{2}, \frac{1}{2}\right]$, then $x \le g(x) \le \frac{1}{2}$

Let $x_0 \in \left[-\frac{1}{2}, \frac{1}{2}\right]$ and $x_{n+1} = g(x_n)\ \forall\ n \geq 0$. The sequence x_n is increasing and bounded above so its limit exists and this limit is $\frac{1}{2}$ (by solving g(l) = l)

By continuity, the constant sequence $\langle f(x_n) \rangle$ tends to $f\left(\frac{1}{2}\right) \Rightarrow$ f is constant over $\left[-\frac{1}{2}, \frac{1}{2}\right]$

53. 00002.00

Sol. Let $g(x) = e^{x/2} f(x)$
Using Rolle's Theorem, g′, g″, g‴ have at least 4, 3, 2 distinct real zeroes respectively

54. 00000.50

Sol. $$\lim_{n\to\infty} \frac{\left(\frac{k}{n^2}\right)^{\frac{k}{n^2}+1}}{\frac{k}{n^2}} = 1\ \forall\ k \in N \Rightarrow \lim_{n\to\infty} \frac{\sum_{k=1}^{n}\left(\frac{k}{n^2}\right)^{\frac{k}{n^2}+1}}{\sum_{k=1}^{n}\frac{k}{n^2}} = 1 \Rightarrow \lim_{n\to\infty} \sum_{k=1}^{n}\left(\frac{k}{n^2}\right)^{\frac{k}{n^2}+1} = \lim_{n\to\infty} \sum_{k=1}^{n}\frac{k}{n^2} = \frac{1}{2}$$